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You are a Product Analyst investigating customer refund dispute characteristics across transaction types....

Question 1: For disputes involving digital goods (where the product_type begins with "DIG"), what is the average refund amount for disputes initiated from October 1st to October 7th, 2024? This metric helps quantify the financial impact of digital goods disputes.

```
SELECT AVG(refund_amount) AS avg_refund_amount
FROM fct_disputed_transactions
WHERE dispute_initiated_date >= '2024-10-01' AND dispute_initiated_date <= '2024-10-07'
AND product_type LIKE 'DIG%'
```



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Question 2: For disputes in October 2024, first find the top 5 transaction categories by the number of disputes. Among these top 5 categories, identify the category among these with the highest average dispute resolution time. Report both the category name and its average resolution time to target process improvements.

```
WITH ranked_categories AS (  
    SELECT  
        transaction_category,  
        COUNT(*) AS number_dispute,  
        DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank_num  
    FROM fct_disputed_transactions  
    WHERE dispute_initiated_date >= '2024-10-01'  
        AND dispute_initiated_date < '2024-11-01'  
    GROUP BY transaction_category  
)  
  
SELECT  
    t.transaction_category,  
    AVG(t.resolution_time_days) AS avg_resolution_time  
FROM fct_disputed_transactions t  
JOIN ranked_categories r  
    ON t.transaction_category = r.transaction_category  
WHERE t.dispute_initiated_date >= '2024-10-01'  
    AND t.dispute_initiated_date < '2024-11-01'  
    AND r.rank_num <= 5  
GROUP BY t.transaction_category  
ORDER BY 2 DESC  
LIMIT 1
```



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Question 3: Segment all disputes from October 2024 into quartiles based on the resolution time. What percentage of disputes in the highest resolution time quartile involve physical goods (i.e. product_type values not starting with "DIG")? This analysis will guide recommendations to reduce overall dispute resolution time.

```
with high AS(
  SELECT *
  FROM (SELECT product_type,
              resolution_time_days,
              NTILE(4) OVER(ORDER BY resolution_time_days DESC) AS qua
rtiles
  FROM fct_disputed_transactions
  WHERE dispute_initiated_date >= '2024-10-01' AND dispute_ini
tiated_date < '2024-11-01') AS total
  WHERE quartiles = 1)

SELECT ROUND(
  100.0 * SUM(CASE WHEN product_type NOT LIKE 'DIG%' THEN 1
ELSE 0 END) / COUNT(*),
  2
) AS non_digital_percentage
FROM high
```

